TABLE I: Classification report of FAMus for original dataset

Mood	Accuracy	Precision	Recall	F1-score
ANGRY	0.91	0.91	0.91	0.91
CALM	0.9	0.9	0.9	0.9
HAPPY	0.9	0.9	0.9	0.9
NERVOUS	0.89	0.88	0.89	0.88

TABLE II: Feature-group ablation with FedAvg.

Feature set	Accuracy	F1-score
All 19 features	0.9 $\pm$ 0.01	0.9 $\pm$ 0.02
Physiology only	0.9 $\pm$ 0.05	0.9 $\pm$ 0.05
Music only	0.75 $\pm$ 0.01	0.7 $\pm$ 0.02

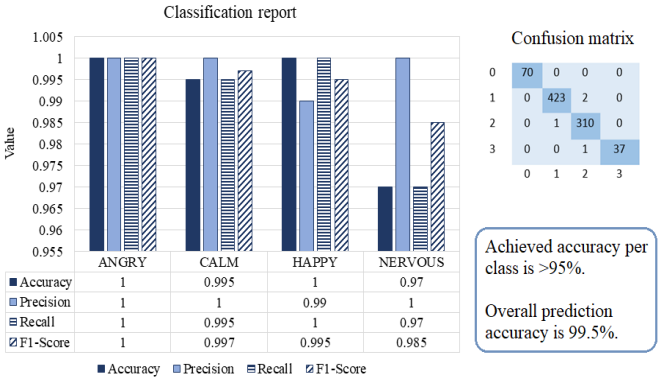


Fig. 1: Classification report of FAMus for CTGAN-based dataset

TABLE III: Prediction accuracy of FAMus and baselines

Model/ Classifier	Original dataset	CTGAN dataset
RF	0.87	0.94
LSTM	0.71	0.8
MLP	0.78	0.8
FAMus with FedAvg	0.9	>0.99
FAMus with FedProx	0.91	>0.99
FAMus with FedNova	0.91	>0.99
FAMus with SCAFFOLD	0.91	>0.99
FAMus with FedMedian	0.9	>0.99

TABLE IV: Response Time in FAMus, Edge-cloud, and Cloud-only frameworks

Approach	Response Time
FAMus	0.079 s
Edge-cloud	0.32 s
Cloud-only	0.78 s

TABLE V: Packet delivery ratio for different communication link

Link	PDR
Ethernet	99%
WiFi	98%
Cellular	95%